

Worked Exercise: Quality Control Across Three Machines

Topics: law of total probability, Bayes rule, random variables, expectation, and model interpretation

Problem. A factory uses three machines. Machine A produces 30% of all items and is 80% accurate; B produces 50% and is 90% accurate; C produces 20% and is 60% accurate. An item is called defective when the machine is not accurate for that item.

- (a) Find the probability that a randomly selected item is defective.
- (b) Given that an item is defective, find the probability that it came from Machine C.
- (c) Let X be the indicator of a defective item: $X=1$ if defective and $X=0$ otherwise. Give the probability mass function of X , $E[X]$, and $\text{Var}(X)$.
- (d) For 100 independently produced items under the same production mix, let Y be the number defective. State a suitable model for Y and compute $E[Y]$.

Step 1 - translate the words into events

Let A , B , C denote the producing machine and D denote a defective item. Then $P(A)=0.30$, $P(B)=0.50$, $P(C)=0.20$. Because accuracy is the probability of a correct item: $P(D|A)=0.20$, $P(D|B)=0.10$, and $P(D|C)=0.40$.

Step 2 - total probability

The machine events form a partition, so every defective item must have come from exactly one of A , B , or C .

$$\begin{aligned} P(D) &= P(D|A)P(A) + P(D|B)P(B) + P(D|C)P(C) \\ &= (0.20)(0.30) + (0.10)(0.50) + (0.40)(0.20) \\ &= 0.06 + 0.05 + 0.08 = 0.19. \end{aligned}$$

Step 3 - reverse the condition using Bayes rule

$$P(C|D) = P(D|C)P(C) / P(D) = (0.40)(0.20)/0.19 = 0.08/0.19 = 0.4211.$$

Interpretation: although C makes only 20% of all items, it is responsible for about 42.1% of defective items because its defect rate is comparatively high.

Step 4 - construct and summarize the random variables

For X , $P(X=1)=0.19$ and $P(X=0)=0.81$. Therefore $E[X]=1(0.19)+0(0.81)=0.19$. Since X is Bernoulli, $\text{Var}(X)=p(1-p)=(0.19)(0.81)=0.1539$.

Assuming independent items with a stable defect probability, $Y \sim \text{Binomial}(n=100, p=0.19)$. Hence $E[Y]=np=100(0.19)=19$ defective items.

Checks. All probabilities lie between 0 and 1; the PMF sums to 1; and the expected count 19 agrees with a 19% defect probability over 100 items.